Statistics of Kerr-thermalized light in multimode fibre: exact Dirichlet mode statistics and the limits of squeezing

Authors: to be set by the PI
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Kerr nonlinearity drives the modal powers of light in a graded-index multimode fibre towards a Rayleigh–Jeans distribution and, at low energy, condenses power into the fundamental mode. Beyond the mean occupations, we collect analytical results on the fluctuations and quantum noise of this light, each checked by an independent simulation written by a second model family. (i) When the excitation energy per unit power is below the first modal gap, the microcanonical Rayleigh–Jeans state has exactly Dirichlet-distributed scaled excitation powers at any mode number M: every excited mode has $g^{(2)} = 2 - 2/M$, and the condensate–mode covariance changes sign at the harmonic mean of the excitation energies. (ii) Before the quantum state scrambles, the multimode quadrature squeezing of a coherent launch is $\sigma_{\max}^{-2}$, from the classical tangent map, only while $\sigma_{\max}^6/n_0 \lesssim 1$; the curvature of the Kerr crescent sets the leading correction. (iii) The condensate's own squeezing is $\lambda_- \simeq \det V/V_P$, with a computable error; it is about −9.5 dB at ten nonlinear lengths in a 15-mode model. (iv) On its early shear branch a Kerr stage squeezes a displaced thermal state of mixedness ν at best to $\frac{3}{2\cdot 4^{1/3}}(\nu^4/n_d)^{1/3}$, and cannot squeeze a phase-invariant state at all. We state each regime, what failed in verification, and what to measure.

I. INTRODUCTION

Kerr nonlinearity in a graded-index (GRIN) multimode fibre redistributes power among the guided modes, and the modal powers relax towards a Rayleigh–Jeans (RJ) distribution. At low energy, power condenses into the fundamental mode, which is one account of beam self-cleaning [1–4]. Weak random coupling within degenerate groups accelerates the relaxation [5, 6]; strong coupling between groups competes with it [7]. Wave-turbulence theory [1, 8, 9] and its quantum extensions [10, 11] predict the mean occupations, and experiments measure them by averaging over pulses or speckle realizations [2–4].

This paper concerns what the means leave out: the fluctuations of the modal powers in the thermalized state (Sec. III), the quantum noise of the light early in the propagation, before its quantum state scrambles (Secs. IV A and IV B), and the squeezing of the condensate after it leaves the fibre (Sec. V). For each result we give the steps that carry the derivation, the regime as bounds on dimensionless groups, and how it was checked.

Each result was proposed by one model family (Claude) and checked by an independent implementation written by another (GPT), against predictions frozen before the runs; several original statements failed and were narrowed. None is formalized or peer reviewed. Table I gives the status; the Appendices give full derivations and the run and verifying agent of every number.

We build on known results: finite-mode microcanonical departures from RJ in bosonic oligomers [12]; classical-field condensate statistics in the canonical ensemble [13]; canonical versus microcanonical condensate fluctuations [14]; the coherent-state Kerr optimum $0.945\, n^{-1/3}$ at fixed photon number [15] (ingredients in Ref. [16]); the Kerr crescent and interferometric number squeezing [17] and one-axis twisting [18]; the invariance of phase-averaged states under Kerr evolution [19]; and linearized propagation of multimode quantum noise in fibres [20]. A conference abstract reports entanglement arising during thermalization [21]; we have not seen a full paper. Bounded literature searches did not find the Dirichlet law and its covariance structure for condensed RJ light, the curvature bound on multimode squeezing in a thermalizing fibre, or the mixedness factor $\nu^{4/3}$; these searches do not establish novelty.

II. MODEL AND SCALES

We use a scalar field in one polarization, with modal amplitudes $A_p(z)$, $p = 0, \ldots, M-1$, normalized so that $I_p = |A_p|^2$ is the modal power:

$$i\partial_z A_p = \varepsilon_p A_p - \gamma \textstyle\sum_{qrs} S_{pqrs} A_q^* A_r A_s. \tag{1}$$

Here $\varepsilon_p \geq 0$ is the propagation-constant offset from the fundamental ($\varepsilon_0 = 0$), S_{pqrs} the real, symmetric Kerr overlap and γ the nonlinear coefficient. A parabolic index profile gives K equidistant groups, $\varepsilon = q\kappa$ with $q = 0, \ldots, K-1$ and degeneracy $q+1$, so $M = K(K+1)/2$. When the Kerr energy is small against the gap, $L_{\text{beat}}/L_{\text{NL}} \ll 1$ with $L_{\text{NL}} = 1/(\gamma N)$ and N the total power, averaging over the beat length leaves a resonant normal form [22] that conserves the power $N = \sum_p I_p$, the excitation energy $E' = \sum_p \varepsilon_p I_p$ and the resonant Kerr energy H_{res}. Real fibres have weak random linear coupling among degenerate modes [5, 6]; it conserves N and E' but not H_{res} or the orbital angular momentum L_z. For the scalar model we take Hermitian Brownian coupling of amplitude σ within each degenerate spatial group (the cited works couple polarizations), with decorrelation length $L_d = L_{\text{NL}}/\sigma^2$.

For quantum noise we quantize each temporal coherence cell of duration τ separately, neglecting the coupling of cells by dispersion and walk-off: $[a_p, a_q^\dagger] = \delta_{pq}$ and $n = N\tau/\hbar\omega$ photons per cell (for example $n \approx 5 \times 10^{10}$ for 10 kW in 1 ps at 1 μm). The mean evolution depends only on γN; quantum fluctuations enter through $1/n$. Quadratures $X_\theta = ae^{-i\theta} + a^\dagger e^{i\theta}$ have vacuum variance 1. The fundamental holds $n_0 = f_0 n$ photons.

The dimensionless groups are the scaled energy $E'/\kappa N$ (equivalently the condensate fraction f_0), the mode number M (or K), L_d/L_{NL}, z/L_{NL}, n and $L_{\text{beat}}/L_{\text{NL}}$. The cross-

family checks used an independent implementation of the resonant model with exact Hermite–Gauss overlaps, $K = 4$–6 ($M = 10$–21), tangent-map and truncated-Wigner (TWA) propagation, and exact one- and three-mode quantum references (Appendix F).

III. DIRICHLET STATISTICS OF THE CONDENSED RJ STATE

Statement. Suppose that the only conserved quantities of the modal powers are N and E', and that the statistics are those of the flat (Liouville) measure on that shell. Let $D = M - 1$ be the number of excited modes and $\kappa = \min_{m\geq 1} \varepsilon_m$ the first gap. If

$$0 < E' \leq \kappa N, \tag{2}$$

that is, if the excitation energy per unit power lies below the first gap, the scaled excitation powers $x_m = \varepsilon_m I_m / E'$ ($m = 1, \ldots, D$) are exactly uniform on the simplex $\sum_m x_m = 1$: $x \sim \mathrm{Dirichlet}(1, \ldots, 1)$, with $I_0 = N - \sum_m I_m$.

Crux. Write $A_p = \sqrt{I_p}\, e^{i\phi_p}$. Because $d^2A_p = \frac{1}{2} dI_p\, d\phi_p$, the measure $\delta(\sum I - N)\, \delta(\sum \varepsilon I - E') \prod_p d^2A_p$ is flat in the intensities on the polytope $\{I \geq 0,\ \sum I = N,\ \sum \varepsilon I = E'\}$. Integrating out I_0 with the first delta function leaves only the condition $I_0 \geq 0$, i.e. $\sum_m I_m \leq N$. Under Eq. (2) this condition cannot bind, since $\sum_m I_m \leq \sum_m (\varepsilon_m/\kappa) I_m = E'/\kappa \leq N$: the condensate absorbs or supplies power at no energy cost. What remains is one linear constraint on D nonnegative variables, and the rescaling to x_m has a constant Jacobian. No large-M or small-depletion expansion enters.

Consequences. The Dirichlet moments $\langle x_m \rangle = 1/D$, $\langle x_m^2 \rangle = 2/[D(D+1)]$ and $\langle x_m x_n \rangle = 1/[D(D+1)]$ ($m \neq n$) give, with $S_r = \sum_{m\geq 1} \varepsilon_m^{-r}$,

$$\langle I_m \rangle = \frac{E'}{D\varepsilon_m}, \quad g_m^{(2)} = 2 - \frac{2}{M}, \quad g_{mn}^{(2)} = 1 - \frac{1}{M}, \tag{3}$$

$$\mathrm{Var}\, I_0 = \frac{E'^2 (D S_2 - S_1^2)}{D^2(D+1)}, \tag{4}$$

$$\mathrm{Cov}(I_0, I_m) = -\frac{E'^2}{\varepsilon_m} \frac{D/\varepsilon_m - S_1}{D^2(D+1)}, \tag{5}$$

where $g_m^{(2)} = \langle I_m^2 \rangle / \langle I_m \rangle^2$. The means are RJ with the chemical potential at the fundamental, exactly, at any M. Every excited mode has the same $g^{(2)}$, independent of its energy and of E'; it falls short of the thermal value 2 by $2/M$, the trace of the fixed total energy. The condensate is anticorrelated with modes below the harmonic mean D/S_1 of the excitation energies and correlated with modes above it. For the GRIN ladder, $D = K(K+1)/2 - 1$, $\kappa S_1 = K - 1 + H_{K-1}$ and $\kappa^2 S_2 = H_{K-1} + H_{K-1}^{(2)}$ (harmonic numbers), so the sign change lies between groups 2 and 3 for $K = 5$ and 6, and the depletion $N_{\mathrm{ex}} = N - I_0$ has $\mathrm{Var}\, N_{\mathrm{ex}} / \langle N_{\mathrm{ex}} \rangle^2 \to (\ln K + \gamma_E + \pi^2/6 - 2)/K^2$ (Appendix A). The single-mode law, $\mathrm{Beta}(1, D-1)$, is the textbook energy law of one of D oscillators at fixed total energy; what is new is that the condensate maps the condensed RJ state exactly onto it. Above $E' = \kappa N$ the condition $I_0 \geq 0$ binds (Appendix A).

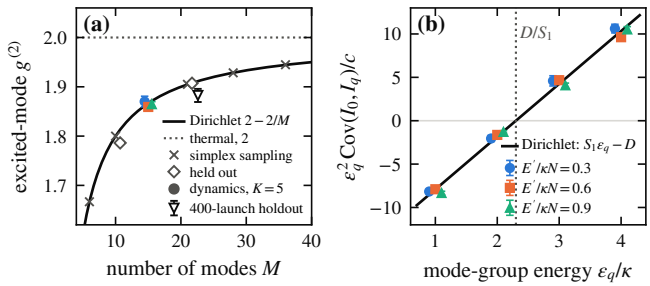


FIG. 1. Dirichlet statistics of condensed RJ light. (a) Excited-mode $g^{(2)}$ against mode number. Line: Eq. (3). Crosses: sampling of the measure. Coloured symbols: dynamics at $K = 5$, $L_d/L_{\mathrm{NL}} = 11.1$, $z/L_{\mathrm{NL}} \in [2000, 8000]$, for $E'/\kappa N = 0.3$ (circles), 0.6 (squares), 0.9 (triangles). Diamonds: held-out $K = 4, 6$. Down-triangle: fresh 400-launch holdout. Error bars: one standard error (SE) over disorder realizations. (b) Condensate–mode covariance for $K = 5$, multiplied by ε_q^2/c with $c = E'^2/[D^2(D+1)]$, so that Eq. (5) becomes the line $S_1\varepsilon_q - D$ ($\kappa = 1$), which crosses zero at $D/S_1 = 2.30\kappa$. Near the crossing relative errors are larger (30% at $q = 2$, $E'/\kappa N = 0.9$); the verified statement is the sign. Run: `2026-10-17-gpt-b-dirichlet`.

Checks. Independent sampling of the simplex for $M = 6$–36 gives $2 - g^{(2)} \propto M^{-1.0017 \pm 0.0021}$ with coefficient 2.008 ± 0.012 (exact: 2). The dynamical tests used $L_d/L_{\mathrm{NL}} = 11.1$, the window $z/L_{\mathrm{NL}} \in [2000, 8000]$ and six launch and disorder realizations per case. At $K = 5$ and $E'/\kappa N = 0.3$, 0.6, 0.9 the pooled excited-mode $g^{(2)}$ is 1.870, 1.859, 1.865 against $28/15 = 1.867$, and $\mathrm{Var}\, I_0$ is 1.03 ± 0.02, 0.99 ± 0.04 and 1.04 ± 0.03 times Eq. (4); every resolved covariance sign agrees with Eq. (5) [Fig. 1]. Held-out sizes pass: $K = 4$ at $E'/\kappa N = 0.45$ and $K = 6$ at 0.75 give $g^{(2)} = 1.786$ and 1.907, against 1.800 and 1.905. Failures are part of the record: the proposer's registered prediction for a toy model at $L_d \approx L_{\mathrm{NL}}$ failed ($g^{(2)} = 1.67$–1.74 against 1.80; the toy was not ergodic within $2 \times 10^4 L_{\mathrm{NL}}$), and a 200-launch fixed-length sample at $K = 6$ failed three single-mode $g^{(2)}$ tolerances. Its 800-launch enlargement (a post-failure diagnostic) and a fresh frozen 400-launch holdout at $E'/\kappa N = 0.5$ ($g^{(2)} = 1.883 \pm 0.013$ against 1.905) pass. At $L_d/L_{\mathrm{NL}} = 0.11$ the case $E'/\kappa N = 0.3$ fails a moment test in that window; all pass for $z/L_{\mathrm{NL}} \in [8000, 24000]$, with variance ratios uncertain by 12–17%.

Not claimed. The flat measure is stationary, but its selection from a given launch requires mixing, tested only at the listed parameters; no equilibration length is claimed. The statement is classical: its reading as photon statistics of Fock-diagonal reduced states after scrambling, and two-polarization extensions, are not verified.

IV. SQUEEZING BEFORE THE STATE SCRAMBLES

A coherent launch evolves unitarily: the light stays pure, single-mode mixedness is entanglement with other modes,

and each cell keeps its Poisson photon-number distribution. Repeated shots of a fixed launch reproduce the same speckle up to quantum noise until the state scrambles. TWA simulations of the 15-mode model place the half-collapse of the mean field at 331–411 $L_{\rm NL}$ for $n = 10^8$ on two disorder paths (halving the step moves one from 331 to 383), growing roughly as $\ln n$; the coefficient is not converged (run `2026-10-17-gpt-b-scrambling`). We ask when the fluctuations about the classical trajectory, of relative size $n^{-1/2}$, are Gaussian and how squeezed they are.

A. The Gaussian window is set by curvature

Exact single-mode Kerr evolution, $\hat{H} = \chi\hat{N}(\hat{N}-1)/2$ with shear $s = \chi n z$, shows that coherence is the wrong clock. The coherence obeys $|\langle a\rangle|^2/n = e^{-4n\sin^2(s/2n)}$, with half-collapse at $s \simeq \sqrt{n\ln 2}$, but at $n = 4096$ and $s = 4$, where the coherence is still 0.9961, the covariance already has $\det V = 2.513$ against $1/4$ for a pure Gaussian state (vacuum variance $1/2$ in this convention; run `2026-10-17-gpt-b-scrambling`). A pure state need not be Gaussian; this refuted the original claim of a pure Gaussian state until coherence collapses, and it is why the results below carry an explicit Gaussian window.

Linear stage. Linearize $a_p = \sqrt{n}A_p(z) + \delta a_p$. The quadrature fluctuations evolve under the real symplectic tangent map $S(z)$ of the classical flow, so a coherent input gives $V = SS^{\mathsf{T}}$. Symplectic singular values come in pairs (σ, σ^{-1}), so the least variance over all multimode quadratures is

$$\lambda_{\rm lin} = \sigma_{\rm max}^{-2}. \tag{6}$$

In the 15-mode model this is −24.7 to −25.7 dB at $z = 10L_{\rm NL}$ and −30.1 to −31.6 dB at $20L_{\rm NL}$ over the tested launches and disorder paths at $E'/\kappa N = 0.3$–0.6 (run `2026-10-18-gpt-b-grin-squeezing`).

Crux of the correction. The long axis of the squeezed ellipse is a phase direction, which Kerr evolution bends along a circle of radius $2\sqrt{n_0}$ in quadrature space. A point displaced by ℓ along the arc leaves the tangent line by the sagitta $\ell^2/(4\sqrt{n_0})$. For a Gaussian spread of variance V' along the arc, the sagitta has variance $V'^2/(8n_0)$ and is uncorrelated with the narrow coordinate, so it adds to the narrow variance. With $V' \simeq \sigma_{\rm max}^2$,

$$\frac{V_{\rm fixed}}{\lambda_{\rm lin}} - 1 \simeq c\,R, \qquad R \equiv \frac{\sigma_{\rm max}^6}{n_0}, \tag{7}$$

with $c = 1/8$ for one mode. $V_{\rm fixed}$ is the true variance along the tangent map's squeezed direction, chosen in advance, and $R \le 0.8$ corresponds to a 10% correction.

Checks. Exact single-mode evolution gives a relative excess of 0.1106 at $n_0 = 10^4$ and 0.1002 at 10^{10} for $R = 0.8$, approaching $R/8$ (run `2026-10-18-gpt-b-kerr-exact`). In 15 modes, nonlinear Wigner ensembles give $c = 0.092 \pm 0.004$ at $E'/\kappa N = 0.6$ and $z = 10L_{\rm NL}$, with occupation exponent -0.995 ± 0.010 over $n = 3\times10^5$–3×10^6 [Fig. 2(a)]; c is a calibration for that trajectory, not a universal constant, and we have no derivation of its value. Within the checked envelope—$10^6 \le n \le 10^{10}$, $5 \le z/L_{\rm NL} \le 20$, $0.55 \le f_0 \le 0.87$, $L_d/L_{\rm NL} = 11.1$ and $R \le 0.8$—the fixed-direction variance stays within 0.37 dB of $\lambda_{\rm lin}$. Frozen held-out cases pass: at $E'/\kappa N = 0.3$ and $n = 10^8$ the excess is 0.30, 6.7 and 28.6 dB at $z/L_{\rm NL} = 10$, 20 and 50. The best quadrature found at $n = 10^{10}$ over 25 sampled distances is −31.5 dB; whether a multimode ceiling exists is open.

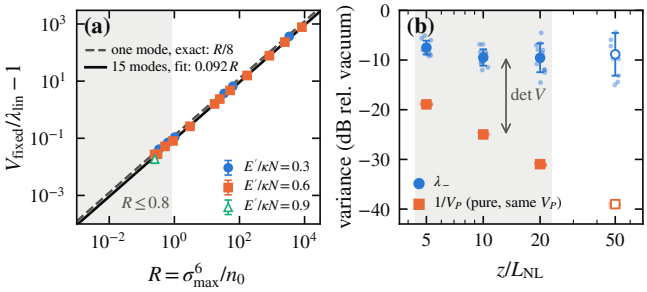


FIG. 2. Squeezing before scrambling in the 15-mode GRIN model ($K = 5$), $L_d/L_{\rm NL} = 11.1$. (a) Excess of the nonlinear (Wigner) variance along the tangent map's squeezed direction over $\lambda_{\rm lin} = \sigma_{\rm max}^{-2}$, for $z/L_{\rm NL} = 5, 10, 20$ and $n = 3\times10^5$–10^{10}. Open triangle: $E'/\kappa N = 0.9$, $z = 20L_{\rm NL}$, $f_0 = 0.50$, outside the verified range. Excesses below two SE are not shown. Solid line: 15-mode calibration at $z = 10L_{\rm NL}$; dashed: exact single mode. Shaded: $R \le 0.8$. Runs: `2026-10-18-gpt-b-grin-squeezing`, `2026-10-18-gpt-b-kerr-exact`. (b) The condensate's least quadrature variance λ_- (circles) and the pure-state value $1/V_P$ (squares), for 12 disorder paths at $E'/\kappa N = 0.3$, $n = 10^{10}$. Large symbols: mean ± SD; dots: single paths. The gap is $\det V$ [Eq. (8)]. Shaded: verified range; open symbols: not resolved under step refinement. Run: `2026-10-18-gpt-b-condensate-squeezing`.

Not established. Curvature immunity of weighted photon counts, argued from their phase insensitivity, holds for pure self-phase modulation but is unproven under intensity exchange: disorder-free checks pass, no continuum counterexample was found, and late-distance tests remain step-size sensitive (run `2026-10-18-gpt-b-grin-squeezing`). Linearized predictions of deeper squeezing at $400L_{\rm NL}$, far outside $R \le 0.8$, were withdrawn.

B. The condensate's own squeezing

Consider the fundamental alone, with covariance $V = \left(\begin{smallmatrix} V_X & C \\ C & V_P \end{smallmatrix}\right)$ in its amplitude and phase quadratures and eigenvalues $\lambda_- \le \lambda_+$.

Crux. Since $\lambda_-\lambda_+ = \det V$ and $\lambda_- + \lambda_+ = V_X + V_P$,

$$\frac{\det V/V_P}{\lambda_-} - 1 = \frac{V_X - \lambda_-}{V_P} \in \left[0, \frac{V_X}{V_P}\right], \tag{8}$$

so for an ellipse elongated along the phase, $\lambda_- \simeq \det V/V_P = V_X V_{P|X}/V_P$ with $V_{P|X} = V_P - C^2/V_X$. The identity needs no Gaussian assumption. Its reading: self-phase modulation makes V_P large; a pure state ($\det V = 1$) would give $\lambda_- \simeq 1/V_P$; phase noise that is not linearly correlated with the condensate's own amplitude, present because the condensate is entangled with the excited modes, raises $\det V \equiv \nu^2$ and

costs $20\log_{10}\nu$ dB. $V_{P|X}$ is a regression residual; it should not be read as a unique cross-phase contribution of the excited modes.

Checks. Over 1177 tested covariances Eq. (8) errs by at most 0.062 dB ($V_X/V_P \le 0.023$). For 12 independent disorder paths at $E'/\kappa N = 0.3$, $L_d/L_{\rm NL} = 11.1$ and $n = 10^{10}$, the condensate is squeezed to −9.49 dB on average at $z = 10L_{\rm NL}$ (SD over paths 1.65 dB; $f_0 = 0.80$–0.86), against −25.0 dB for $1/V_P$ [Fig. 2(b)]. At $20L_{\rm NL}$ the mean is −9.55 dB (SD 2.91 dB), but four paths exceed $f_0 = 0.87$; the eight in range give −7.95 dB (SD 1.98 dB). The curvature ratio $V_P^2/(8n_0\lambda_-)$ stays below 10^{-3}, and frozen nonlinear-Wigner ensembles agree with the tangent values within 0.18 dB through $20L_{\rm NL}$. The spread over paths does not average out. Stronger disorder ($L_d/L_{\rm NL} = 0.25$, 0.11) fails step-resolution gates at $20L_{\rm NL}$, and at $50L_{\rm NL}$ single paths still change by 1.3 dB under refinement, so the verified range stops at $20L_{\rm NL}$ (run `2026-10-18-gpt-b-condensate-squeezing`).

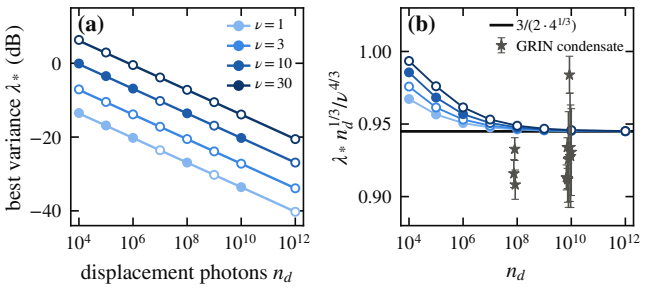


FIG. 3. Kerr squeezing of a displaced thermal state, $V = \nu I$. (a) Exact optimum λ_* against displacement photons n_d; lines: Eq. (9); open symbols: held out of the run's fits ($\nu = 3$, 30; $n_d = 10^7, 10^9, 10^{12}$). (b) The same divided by $\nu^{4/3}n_d^{-1/3}$ (colours as in (a)); it approaches $3/(2\cdot 4^{1/3})$ with relative error decaying as $n_d^{-1/3}$. Stars: Kerr stage applied to the condensate extracted from the 15-mode simulations (nonlinear Wigner input, $n_d = n_0$, ν from its tangent covariance; error bars: SE over vacuum batches; spread horizontally for visibility). Runs: `2026-10-18-gpt-b-kerr-exact`, `2026-10-18-gpt-b-condensate-squeezing`.

V. SQUEEZING THE EXTRACTED CONDENSATE

Suppose the fundamental is separated after the fibre and sent through a lossless single-mode Kerr stage, $\hat{H} = \chi\hat{N}(\hat{N}-1)/2$, with shear $s = \chi n_d z$ (the stage's nonlinear phase). The verified statement concerns a displaced thermal input, with covariance $V = \nu I$ and $n_d = |\alpha|^2$ displacement photons.

Crux. Linearized, the stage is the shear $P \to P + 2sX$. It squeezes νI to $\nu/(4s^2)$ for $s \gg 1$ while stretching the phase direction to $V' \simeq 4\nu s^2$, so the curvature term of Sec. IV A adds $V'^2/(8n_d) = 2\nu^2 s^4/n_d$. Minimizing $\lambda(s) = \nu/(4s^2) + 2\nu^2 s^4/n_d$ over s^2 on the early shear branch, $s = O(n_d^{1/6})$, gives $s_*^6 = n_d/(16\nu)$ and

$$\lambda_* = \frac{3}{2\cdot 4^{1/3}}\left(\frac{\nu^4}{n_d}\right)^{1/3}\left[1 + O_\nu(n_d^{-1/3})\right], \tag{9}$$

with $3/(2\cdot 4^{1/3}) = 0.945$. The exponent 1/3 comes from balancing s^{-2} against s^4/n_d, and the factor $\nu^{4/3}$ from the linear term scaling as ν and the curvature term as ν^2. At the optimum the curvature term is half the linear one, outside any Gaussian window, so the balance must be checked against exact evolution. For $\nu = 1$ it reproduces the known coherent-state optimum [15].

Checks. Exact moments (coherent-state Kerr moments averaged over the input's Gaussian P function) agree with Fock evolution to 4×10^{-12}, and their expansion gives Eq. (9) (run `2026-10-18-gpt-b-kerr-exact`). Over $1 \le \nu \le 30$ and $10^8 \le n_d \le 10^{12}$ the coefficient is within 0.41% of 0.945; at $\nu = 10$ the measured exponent is −0.3346 and the relative error decays as $n_d^{-0.328}$; all nine frozen held-out cases ($\nu = 3, 10, 30$; $n_d = 10^4$–10^6) are within 0.22 dB [Fig. 3]. For the condensate extracted from the 15-mode simulations the law matches 12 frozen cases ($n = 10^8$, 10^{10}; $z/L_{\rm NL} = 10$–50; two energies and disorder strengths) within 0.18 dB, with phase and angle cross-validated on separate vacuum batches; at $z = 10L_{\rm NL}$, $n = 10^{10}$: $\nu = 5.7$, $\lambda_* = -23.4$ dB (run `2026-10-18-gpt-b-condensate-squeezing`). The mixedness alone does not fix λ_* for an arbitrary Gaussian input: an oriented, squeezed input can beat Eq. (9).

No-go for phase-invariant states. If $[\rho, \hat{N}] = 0$, every number-diagonal (Kerr) unitary leaves ρ unchanged, $\langle a\rangle = \langle a^2\rangle = 0$, and

$$\mathrm{Var}\, X_\theta = \langle aa^\dagger + a^\dagger a\rangle = 1 + 2\langle\hat{N}\rangle \ge 1 \tag{10}$$

at every angle [19]. Averaging the preparation over the global input phase is sufficient, because number-conserving evolution keeps the global phase invariance and the partial trace preserves it. Loss of first-order coherence alone is not sufficient. The theorem is single-mode; it does not exclude two-mode, number-conserving readouts.

Self-referenced Sagnac squeezer. In a balanced Sagnac loop the pump's phase noise and mixedness leave through the bright port, and the dark port carries vacuum sheared by the mean field of each arm. A relative shot-to-shot power jitter ε changes the shear Φ and tilts the optimal angle by $\delta\psi \simeq \varepsilon/(2\Phi)$; times the anti-squeezed variance $4\Phi^2$, this adds ε^2 to the dark-port variance, to leading order in ε and $1/\Phi$. With pump powers drawn from the Dirichlet law of Sec. III ($K = 5$, $E'/\kappa N = 0.6$, $\varepsilon^2 = 2.7\times 10^{-3}$), the exact two-arm calculation with the bright port as quantum reference gives −25.4 dB at $\Phi = 30$ and 10^{10} photons per arm, against −35.6 dB without jitter; baseline plus ε^2 misses the jittered value by 3.6% (run `2026-10-18-gpt-b-kerr-exact`). The ε^2 term is a leading small-jitter result for a specified readout, not a finite-error bound: with raw shot-noise normalization a 1% accuracy claim fails at $\varepsilon = 0.051$, $\Phi = 3$, and an external phase-tracking oscillator has a different zero-jitter baseline (−33.6 dB). A prepared Dirichlet ensemble of pump powers is not the photon-number distribution of a single thermalized output.

TABLE I. Status of each statement (October 18, 2026). All were proposed by claude-interactive.7 (Claude) and checked by gpt-b (GPT) with an independent implementation; "sim-verified" is the repository's status for a claim confirmed by simulation of another model family, below "audited" and "formalized". Claim ids: C16-0k is C-20261016-claude-interactive.7-0k and C18-0k is C-20261018-claude-interactive.7-0k.

Statement	Status	Verified regime	Narrowed or failed
Dirichlet law, Eqs. (3)–(5) [C16-01]	sim-verified, narrowed	Measure identity exact for $0 < E'/\kappa N \le 1$, one ground mode. Dynamics: K = 4–6, $L_d/L_{\rm NL} = 11.1$, $z/L_{\rm NL} \in [2000, 8000]$	Classical, scalar only; toy prediction at $L_d \approx L_{\rm NL}$ and a 200-launch sample failed; quantum reading unverified
Pure Gaussian state until coherence collapse [C16-02]	refuted	Exact one-mode collapse formula survives	Gaussianity lost at $s \sim n^{1/6}$; $\ln n$ scrambling trend partly supported, not converged
Early multimode squeezing, Eqs. (6), (7) [C18-01]	sim-verified, narrowed	$10^6 \le n \le 10^{10}$, $5 \le z/L_{\rm NL} \le 20$, $0.55 \le f_0 \le 0.87$, $L_d/L_{\rm NL} = 11.1$, $R \le 0.8$; 15 modes	c trajectory-specific; exact Gaussianity refuted; counting immunity unproven
Condensate identity Eq. (8) and its squeezing [C18-02]	sim-verified, narrowed	Identity exact. Dynamics: $n = 10^{10}$, $5 \le z/L_{\rm NL} \le 20$, $0.6 \le f_0 \le 0.87$, $L_d/L_{\rm NL} = 11.1$, $V_P^2/(8n_0\lambda_-) \le 0.1$	Not a unique cloud term; no universal spread; strong disorder unresolved; a spread prediction failed
Kerr-stage optimum Eq. (9) [C18-03]	sim-verified, narrowed	Displaced thermal, early shear branch; $1 \le \nu \le 30$, $10^8 \le n_d \le 10^{12}$ (held out: 10^4–10^6)	Generic Gaussian inputs excluded (counterexample). Coherent case is prior art
No-go Eq. (10) [C18-03]	exact (proof checked)	Any single-mode state with $[\rho, \hat{N}] = 0$	Scrambling does not imply $[\rho, \hat{N}] = 0$
Sagnac ε^2 term [C18-03]	sim-verified, narrowed	$\varepsilon \ll 1$, $\Phi \gg 1$, $\Phi^2/n_a \ll 1$; specified readout	1% finite-error bound fails with raw normalization
Generalized-Gibbs memory [C16-03]	not verified	—	Phase memory reproduced; aligned-launch endpoint missed (0.569 vs 0.581); mixing unproven

VI. STATUS AND LIMITS

Common assumptions of the statements in Table I: a scalar field in one polarization, the resonant normal form with exactly equidistant groups, monochromatic cells without dispersion or walk-off, electronic Kerr only (no Raman), no loss unless stated, ideal detection, at most 21 modes. The disorder at which the dynamical statements were verified, $L_d/L_{\rm NL} = 11.1$, is weaker than in published simulations of real fibres, which use L_d of about 2–40 cm against $L_{\rm NL}$ of order 1 m [5, 6]; at strong disorder the Dirichlet moments need much longer windows and the condensate-squeezing statistics are not converged. Loss acts on any quadrature variance as $\lambda \mapsto 1 - \eta + \eta\lambda$: 90% output transmission and 90% detection give $\lambda_{\rm det} = 0.19 + 0.81\lambda$, a ceiling of −7.2 dB (run `2026-10-18-gpt-b-grin-squeezing`). TWA [23, 24] was calibrated only against exact one- and three-mode references.

Without intra-group disorder the endpoint is conjectured to be the microcanonical ensemble conditioned also on $H_{\rm res}$ and L_z, which remembers input phases; Gibbs measures of $H_{\rm res}$ are invariant [25] and L_z-conserving ensembles are known [26]. Phase memory and the failure of RJ were reproduced up to $1.2 \times 10^6 L_{\rm NL}$ at $K = 5$, but the predicted endpoint missed for an aligned launch (0.569 against 0.581) and mixing is not established (run `2026-10-17-gpt-b-gge`). Also open: the quantum state after scrambling, the scrambling length, a multimode curvature ceiling and late-distance counting squeezing.

VII. WHAT TO MEASURE

Mode statistics. The Dirichlet predictions need per-realization modal decompositions with the input phases redrawn on every shot, in the condensed regime (2); for the GRIN ladder this means $f_0 \ge 1 - \kappa S_1/D$ in the Dirichlet mean ($f_0 \ge 0.57$ at $K = 5$). They have no free parameter: $g^{(2)} = 2 - 2/M$ in every excited mode, cross-correlations $1 - 1/M$, Eq. (4), and the sign change of $\mathrm{Cov}(I_0, I_q)$ at $\varepsilon_q = D/S_1$. Separating them from thermal statistics requires the pooled excited-mode $g^{(2)}$ to better than $2/M$; per-realization decompositions exist (2×300 in a 45-mode fibre [4]). A fixed coherent launch shows these statistics across launch or disorder realizations, not from shot to shot, until its quantum state scrambles.

Squeezing. Early multimode squeezing needs mode-resolved homodyne detection with local oscillators shaped to the tangent map's squeezed mode and locked to the mean field, at $z/L_{\rm NL} \lesssim 20$ and $n \gtrsim 10^{10}$ ($R \le 0.8$). The condensate's own squeezing needs a local oscillator derived from the condensate, because the fibre turns laser power jitter into condensate phase jitter. A Kerr stage on the extracted condensate needs a nonlinear phase near $s_* = (n_d/16\nu)^{1/6}$, about 20–30 rad at $n_d = 10^{10}$ for $1 \le \nu \le 10$; a balanced Sagnac loop referenced to its bright port rejects the pump's phase noise and mixedness and leaves the ε^2 term. Raman, guided-acoustic, excess pump and loss noise in such stages [27, 28] are outside this analysis.

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Appendix A: Dirichlet law: full derivation

Measure. Take one ground mode with $\varepsilon_0 = 0$ and D excited modes with $\varepsilon_m \geq \kappa > 0$. The flat measure on the shell of fixed N and E' is

$$d\mu \propto \delta\Big(\sum_p I_p - N\Big)\,\delta\Big(\sum_m \varepsilon_m I_m - E'\Big)\prod_{p=0}^{D} d^2A_p. \quad \text{(A1)}$$

In polar variables $A_p = \sqrt{I_p}e^{i\phi_p}$, $d^2A_p = d(\mathrm{Re}A_p)\,d(\mathrm{Im}A_p) = r_p\,dr_p\,d\phi_p = \frac{1}{2}dI_p\,d\phi_p$, so the phases are uniform and independent of the intensities, and the intensity density is constant on the polytope $\{I_p \geq 0\}$ cut by the two hyperplanes. Integrating over $I_0 \geq 0$ against the first delta function sets $I_0 = N - \sum_m I_m$ and restricts the excited intensities to $\{I_m \geq 0,\ \sum_m \varepsilon_m I_m = E',\ \sum_m I_m \leq N\}$. If $E' \leq \kappa N$, the last inequality follows from the energy constraint, $\sum_m I_m \leq \kappa^{-1}\sum_m \varepsilon_m I_m \leq N$, and drops out. With $x_m = \varepsilon_m I_m/E'$ the remaining density is constant on $\{x_m \geq 0,\ \sum_m x_m = 1\}$; the Jacobian $\prod_m (E'/\varepsilon_m)$ is constant. This is Dirichlet(1^D), with density $(D-1)!$ on the simplex.

Moments. For Dirichlet(1^D), $\langle\prod_m x_m^{k_m}\rangle = (D-1)!\prod_m k_m!/(D-1+\sum_m k_m)!$. Hence $\langle x_m^k\rangle = k!(D-1)!/(D+k-1)!$, $g_m^{(k)} = \langle x_m^k\rangle/\langle x_m\rangle^k = k!\,D^k(D-1)!/(D+k-1)!$, and in particular $g_m^{(2)} = 2D/(D+1) = 2 - 2/M$ and $g_{mn}^{(2)} = D/(D+1) = 1-1/M$. The single-component marginal is Beta$(1, D-1)$, $P(I_m) = (D-1)(\varepsilon_m/E')(1-\varepsilon_m I_m/E')^{D-2}$ on $[0, E'/\varepsilon_m]$, which tends to the exponential law as $D \to \infty$. The covariances are $\mathrm{Var}\,x_m = (D-1)/[D^2(D+1)]$ and $\mathrm{Cov}(x_m, x_n) = -1/[D^2(D+1)]$.

Condensate. At fixed N, $I_0 = N - N_{\mathrm{ex}}$ with $N_{\mathrm{ex}} = E'\sum_m x_m/\varepsilon_m$. Then

$$\mathrm{Var}\,I_0 = E'^2\Big[S_2\frac{D-1}{D^2(D+1)} - \frac{S_1^2 - S_2}{D^2(D+1)}\Big] = \frac{E'^2(DS_2 - S_1^2)}{D^2(D+1)}, \quad \text{(A2)}$$

$$\mathrm{Cov}(I_0, I_m) = -\frac{E'^2}{\varepsilon_m}\sum_n \frac{\mathrm{Cov}(x_n, x_m)}{\varepsilon_n} = -\frac{E'^2}{\varepsilon_m}\frac{D/\varepsilon_m - S_1}{D^2(D+1)}. \quad \text{(A3)}$$

By Cauchy–Schwarz, $DS_2 \geq S_1^2$, with equality only for equal gaps: then the condensate does not fluctuate although each excited mode does. The mean is $\langle I_0\rangle = N - E'S_1/D$; the density of N_{ex} is the B-spline with knots E'/ε_m (the density of a linear functional of a uniform simplex point).

GRIN sums. Group q ($1 \leq q \leq K-1$) has $q+1$ modes at $\varepsilon = q\kappa$. Then $D = \sum_{q=1}^{K-1}(q+1) = K(K+1)/2 - 1$, $\kappa S_1 = \sum_q (1 + 1/q) = K - 1 + H_{K-1}$ and $\kappa^2 S_2 = \sum_q (1/q + 1/q^2) = H_{K-1} + H_{K-1}^{(2)}$. For $K = 5$: $D = 14$, $\kappa S_1 = 73/12$, $D/S_1 = 2.30\kappa$, $g^{(2)} = 28/15$. For $K = 6$: $D = 20$, $D/S_1 = 2.75\kappa$, $g^{(2)} = 40/21$. For large K, $\mathrm{Var}\,N_{\mathrm{ex}}/\langle N_{\mathrm{ex}}\rangle^2 = DS_2/[(D+1)S_1^2] - 1/(D+1)$, and with $S_1 \simeq K$, $S_2 \simeq \ln K + \gamma_E + \pi^2/6$ and $1/(D+1) \simeq 2/K^2$ this tends to $(\ln K + \gamma_E + \pi^2/6 - 2)/K^2$, with relative corrections $O(\ln K/K)$. The approach is slow: the exact finite-K expression is 0.49 of the limit at $K = 10$ (0.0125 against 0.0252) and 0.89 at $K = 100$. The form matches the canonical-ensemble result for the two-dimensional ideal Bose gas with the cutoff K in place of $k_BT/\hbar\omega$ [14]; this is a consistency check, not prior art for the cutoff case.

Beyond the condensed bound. For $E' > \kappa N$ the condition $\sum_m I_m \leq N$ cuts the simplex. In general the uniform measure on the full polytope $\{I \geq 0, \sum I = N, \sum \varepsilon I = E\}$ has marginals given by B-splines (Curry–Schoenberg); this was derived and checked by Monte Carlo in the proposer family only. A cross-family attack at $E'/\kappa N = 1.2$, $K = 3$ confirmed that the Dirichlet values fail there ($g^{(2)} = 1.595$–1.616 against $5/3$; run `2026-10-17-gpt-b-dirichlet`).

Fluctuating invariants. For an input whose N and E' fluctuate, the statistics are a mixture over shells and must be computed with the law of total covariance. Adding $\mathrm{Var}\,N$ to Eq. (4) is correct only when E' is sharp and every sampled N satisfies Eq. (2). At $E' = 0$ the normalized excited-mode correlations are undefined.

Two polarizations (partly checked). With d_0 degenerate ground modes and only N, E' conserved, integrating out the ground modes leaves a factor $(N - N_{\mathrm{ex}})^{d_0-1}$, which tilts the Dirichlet law with relative corrections $O(N_{\mathrm{ex}}/N)$. A further conserved quantity per extra ground mode restores the exact form. gpt-b's sampling of the $d_0 = 2$ measure confirmed that the scalar formula fails increasingly with excited power (run `2026-10-17-gpt-b-dirichlet`); the exact spin-constrained form was derived in the proposer family and not cross-checked.

Appendix B: Tangent map and the curvature correction

Linearization. Write $a_p = \sqrt{n}\,A_p(z) + \delta a_p$, with $A_p(z)$ the classical solution of Eq. (1) normalized to $\sum |A_p|^2 = 1$. To first order, δa obeys a linear Bogoliubov equation whose real form, $\xi = (\delta X_1, \ldots, \delta X_M, \delta P_1, \ldots, \delta P_M)$, is $\xi(z) = S(z)\xi(0)$ with $SJS^{\mathsf{T}} = J$. A coherent input has $\langle\xi\xi^{\mathsf{T}}\rangle = I$, so $V = SS^{\mathsf{T}}$, independent of n. Because $S^{-1} = J^{\mathsf{T}}S^{\mathsf{T}}J$, the spectrum of $S^{\mathsf{T}}S$ is closed under inversion, the singular values come in pairs $(\sigma, 1/\sigma)$, and $\lambda_{\min}(V) = \sigma_{\min}^2 = \sigma_{\max}^{-2}$. The linearization is valid while the quadratic terms neglected in δa are small, which Eq. (7) quantifies.

One mode. For $\hat{H} = \chi\hat{N}(\hat{N}-1)/2$ and a coherent amplitude $\sqrt{n_0}$ (real), an amplitude fluctuation δX changes the photon number by $\sqrt{n_0}\,\delta X$ and the accumulated phase by $-\chi z\sqrt{n_0}\,\delta X$; since $\delta P = 2\sqrt{n_0}\,\delta\phi$, the map is the shear $\delta P \to \delta P - 2s\,\delta X$ with $s = \chi n_0 z$ (the sign depends on conventions and does not affect eigenvalues; Sec. V writes $+2s$). Its covariance $\left(\begin{smallmatrix} 1 & -2s \\ -2s & 1+4s^2 \end{smallmatrix}\right)$ has eigenvalues $\lambda_\pm = (\sqrt{1+s^2} \pm s)^2$, so $\lambda_- \simeq 1/(4s^2)$ and $\sigma_{\max}^2 = \lambda_+ \simeq 4s^2$ for $s \gg 1$.

Curvature. In the full evolution each amplitude rotates by its own phase, so the sheared ellipse lies along an arc of radius $r = 2\sqrt{n_0}$ (vacuum variance 1). A point at arc length ℓ from the mean deviates from the tangent line by $h = \ell^2/(2r) + O(\ell^4/r^3)$. If ℓ is Gaussian with variance V', then $\mathrm{Var}\,h = \mathrm{Var}(\ell^2)/(4r^2) = V'^2/(8n_0)$, and h is uncorrelated with the narrow coordinate u because $\langle\ell^2 u\rangle - \langle\ell^2\rangle\langle u\rangle$ vanishes for zero-mean jointly Gaussian (ℓ, u). The tangent squeezed

direction is tilted from the radial direction by $O(1/s)$, which changes h's projection only at higher order. Hence $V_{\rm fixed} \simeq \lambda_{\rm lin} + V'^2/(8n_0)$. With $V' \simeq 4s^2 = \sigma_{\rm max}^2$ and $\lambda_{\rm lin} = \sigma_{\rm max}^{-2}$, $V_{\rm fixed}/\lambda_{\rm lin} - 1 \simeq \sigma_{\rm max}^6/(8n_0)$. The exact single-mode values at $R = 0.08$, 0.8, 8 and 80 approach $R/8$ with corrections that shrink with n_0 (run `2026-10-18-gpt-b-kerr-exact`, Table III).

Many modes. We use n_0 as the curvature scale because the largest shear is the condensate's self-phase modulation. The squeezed direction is a multimode combination, and the measured coefficient, $c = 0.092$, is smaller than $1/8$; we have no derivation of its value or of its dependence on the trajectory. The measured exponent -0.995 ± 0.010 in n at fixed z tests the $1/n_0$ scaling directly.

Appendix C: The condensate covariance identity

For a real symmetric positive 2×2 matrix V with eigenvalues $\lambda_- \le \lambda_+$, $\lambda_-\lambda_+ = \det V$ and $\lambda_- + \lambda_+ = V_X + V_P$. Thus $\det V/V_P = \lambda_-(V_X + V_P - \lambda_-)/V_P = \lambda_-[1 + (V_X - \lambda_-)/V_P]$, which is Eq. (8). Since $0 \le \lambda_- \le \min(V_X, V_P)$, the relative error lies in $[0, V_X/V_P]$: the approximation always overestimates λ_-. Writing $\det V = V_X(V_P - C^2/V_X)$ exhibits the Schur complement $V_{P|X}$, the variance of P left after the best linear prediction from X. It is not the phase variance after a measurement of X, which does not commute with P. For a Gaussian state $\det V = \nu^2$ with ν the symplectic eigenvalue; for a non-Gaussian state the identity holds but ν loses its meaning as a purity. A single-mode reference with the condensate's accumulated self-phase $\Phi_{\rm self}$ has $V_P = 1 + 4\Phi_{\rm self}^2$; its ratio to the computed V_P averages 0.82, 0.86 and 0.89 at $z/L_{\rm NL} = 10$, 20 and 50 (run `2026-10-18-gpt-b-condensate-squeezing`). The ratio is a reference, not an additive causal fraction: switching terms off changes the trajectory.

Appendix D: Kerr stage on a displaced thermal state

Leading balance. The input $V = \nu I$ is the displaced thermal state with $t = (\nu-1)/2$ thermal photons on top of n_d displacement photons. Linearized, the stage gives $V(s) = \nu K_s K_s^{\mathsf T}$ with $K_s = \left(\begin{smallmatrix} 1 & 0 \\ 2s & 1 \end{smallmatrix}\right)$, so $\lambda_{\rm lin} = \nu(\sqrt{1+s^2} - s)^2 \simeq \nu/(4s^2)$ and $V' \simeq 4\nu s^2$. Adding the curvature term $V'^2/(8n_d) = 2\nu^2 s^4/n_d$ and writing $u = s^2$, $\lambda(u) = \nu/(4u) + 2\nu^2 u^2/n_d$. Then $\lambda'(u) = 0$ gives $u_*^3 = n_d/(16\nu)$; at u_* the curvature term equals $\nu/(8u_*)$, half the linear term, and $\lambda_* = 3\nu/(8u_*) = \frac{3}{8}(16\nu/n_d)^{1/3}\nu = \frac{3}{2\cdot 4^{1/3}}(\nu^4/n_d)^{1/3}$. In the scaling $s = c\, n_d^{1/6}$ both terms are $O(n_d^{-1/3})$: $\lambda = n_d^{-1/3}[\nu/(4c^2) + 2\nu^2 c^4] + O(n_d^{-2/3})$.

Exact moments used for verification. With $\tau = \chi z$, $u_r = e^{-ir\tau} - 1$ and $D_r = 1 - tu_r$, averaging the exact coherent-state moments [16] over the Gaussian P function of the input gives

$$\langle a^r\rangle = \alpha^r e^{-i\tau r(r-1)/2} D_r^{-r-1} \exp(n_d u_r/D_r), \qquad r = 1, 2, \tag{D1}$$

and the optimized variance is $\lambda = 1 + 2(n_d + t - |\langle a\rangle|^2) - 2|\langle a^2\rangle - \langle a\rangle^2|$. For $t = 0$ this reduces to the familiar coherent result $|\langle a\rangle|^2 = n_d e^{-4n_d \sin^2(\tau/2)}$. The verification evaluated these expressions at 60-digit precision, compared them with Fock-basis evolution, and expanded them for $s = c n_d^{1/6}$ to obtain Eq. (9); replacing n_d by the total mean photon number $n_d + t$ changes λ_* by a relative $O(\nu/n_d)$ (run `2026-10-18-gpt-b-kerr-exact`).

Why mixedness alone is not enough. A Gaussian input $V = \mathrm{diag}(10^4, 10^{-4})$ at $n_d = 10^{10}$ has $\nu = 1$ and already has variance 10^{-4}, below the coherent optimum 4.39×10^{-4}, with no Kerr stage. Hence Eq. (9) is a statement about displaced thermal inputs (and, numerically, about the extracted condensates of Fig. 3), not a bound for all Gaussian states with a given ν.

Appendix E: No-go theorem and the Sagnac term

No-go. For a single mode, $[\rho, \hat N] = 0$ means $\rho = \sum_n p_n |n\rangle\langle n|$. Any $U = f(\hat N)$ commutes with ρ. Since a and a^2 lower the photon number, $\langle a\rangle = \langle a^2\rangle = 0$, and $\langle X_\theta^2\rangle = \langle a^2\rangle e^{-2i\theta} + \text{c.c.} + \langle aa^\dagger + a^\dagger a\rangle = 1 + 2\langle\hat N\rangle$. Global phase averaging of the input is sufficient: $e^{i\phi\hat N_{\rm tot}}$ commutes with any number-conserving evolution, so the averaged state stays invariant, and the partial trace over the other modes of a state invariant under $e^{i\phi\hat N_{\rm tot}}$ is invariant under $e^{i\phi\hat N_0}$. Checks: Poisson, thermal and nonclassical number mixtures give $\mathrm{Var}\, X_\theta = 1 + 2\langle\hat N\rangle$ to numerical precision (run `2026-10-18-gpt-b-kerr-exact`). A related statement in the original claim, that a phase-randomized state with a smooth super-Poissonian number distribution is a mixture of coherent states, is false in general: a full-support, unimodal mixture of a thermal and a photon-added thermal distribution (mean 20.89, variance 220.10) violates a necessary condition for a nonnegative Glauber–Sudarshan P function (same run).

Sagnac. Let the per-shot shear be $\Phi(1+\delta)$ with $\langle\delta\rangle = 0$, $\langle\delta^2\rangle = \varepsilon^2$, and fix the measurement direction u at the minor axis of $V(\Phi) = \left(\begin{smallmatrix} 1 & 2\Phi \\ 2\Phi & 1+4\Phi^2 \end{smallmatrix}\right)$. Averaging $u^{\mathsf T}V(\Phi(1+\delta))u$ over δ, the linear term vanishes and the quadratic term gives $\frac{1}{2}\Phi^2\varepsilon^2\, u^{\mathsf T}\partial_\Phi^2 V u = 4\Phi^2\varepsilon^2 u_P^2$. With $u_P^2 = \frac{1}{2}(1 - \Phi/\sqrt{1+\Phi^2})$ the added variance is $2\Phi^2(1 - \Phi/\sqrt{1+\Phi^2})\varepsilon^2 = \varepsilon^2[1 + O(\Phi^{-2})]$, which is the leading term quoted in Sec. V for a tracking local oscillator. The bright-port counting readout and the raw versus per-shot normalization differ at the next order; the numbers in Table III specify which readout each refers to.

Appendix F: Numerical methods of the cross-family checks

The checks were written by gpt-b without access to the proposer's code or notes. The resonant model uses exact Hermite–Gauss overlaps (Gauss–Hermite quadrature of exact degree, normalized to $S_{0000} = 1$), complete degenerate groups $q = n_x + n_y < K$, and the resonance mask $q_p + q_q = q_r + q_s$. Disorder is a full Hermitian Brownian matrix within each group, with covariance $\langle dB_{ij}\, dB_{kl}\rangle = \delta_{il}\delta_{jk}\, dz/(q+1)$ times σ^2, applied as unitary steps between RK4 nonlinear steps, so that each step conserves N and E'; we use $L_d/L_{\rm NL} = \sigma^{-2}$.

TABLE II. Provenance. Every number in the text, with its run (folder under `runs/`) and the agent that produced it: the verifier gpt-b, except the failed proposer prediction (run T, claude-interactive.7). Runs: [D] `2026-10-17-gpt-b-dirichlet`; [S] `2026-10-17-gpt-b-scrambling`; [G] `2026-10-17-gpt-b-gge`; [Q] `2026-10-18-gpt-b-grin-squeezing`; [C] `2026-10-18-gpt-b-condensate-squeezing`; [K] `2026-10-18-gpt-b-kerr-exact`; [T] `2026-10-16-claude-interactive.7-rj-toy-thermalization`. Values marked "fig." were computed by `make_figures.py` from the per-path data of the cited run. Closed-form evaluations of displayed formulas (e.g. $28/15$, D/S_1, s_*) and the unit conversion in Sec. II are not from runs.

Quantity	Value	Run	Agent
Dirichlet statistics			
Finite-M exponent of $2 - g^{(2)}$, simplex sampling, $M = 6$–36	1.0017 ± 0.0021	D	gpt-b
Coefficient of $1/M$ (exact 2)	2.0078 ± 0.0123	D	gpt-b
Excited-mode $g^{(2)}$, $K = 5$, $E'/\kappa N = 0.3, 0.6, 0.9$, $L_d/L_{\mathrm{NL}} = 11.1$, $z/L_{\mathrm{NL}} \in [2000, 8000]$	1.8701, 1.8591, 1.8647	D	gpt-b
Var I_0 / Eq. (4), same cases	1.034 ± 0.020, 0.988 ± 0.040, 1.036 ± 0.025	D	gpt-b
Held out: $g^{(2)}$ at $K = 4$ ($E'/\kappa N = 0.45$), $K = 6$ (0.75); variance ratios	1.7862, 1.9072; 0.949 ± 0.039, 1.032 ± 0.047	D	gpt-b
$L_d/L_{\mathrm{NL}} = 0.11$, window $[8000, 24000]$: variance ratios	0.885 ± 0.169, 0.986 ± 0.159, 0.867 ± 0.123	D	gpt-b
200-launch sample, $K = 6$, $E'/\kappa N = 0.75$, $z = 2000 L_{\mathrm{NL}}$: mean $g^{(2)}$; outcome	1.916 ± 0.021; three single-mode tests fail	D	gpt-b
Fresh 400-launch holdout, $K = 6$, $E'/\kappa N = 0.5$, $z = 8000 L_{\mathrm{NL}}$: $g^{(2)}$; variance ratio	1.8825 ± 0.0133; 0.934 ± 0.097	D	gpt-b
Full versus resonant model: error exponent in $(\kappa L_{\mathrm{NL}})^{-1} \in [0.0125, 0.1]$, $z \le 4 L_{\mathrm{NL}}$	0.98–0.99	D	gpt-b
Group-2 condensate covariance at $K = 5$, $E'/\kappa N = 0.9$, relative to Eq. (5)	about 30% less negative (fig.: -1.29 vs -1.83 scaled)	D	gpt-b
Non-condensed attack, $E'/\kappa N = 1.2$, $K = 3$: excited $g^{(2)}$ (Dirichlet 5/3)	1.595–1.616	D	gpt-b
$L_d/L_{\mathrm{NL}} = 0.11$, window $[2000, 8000]$: moment tests at $E'/\kappa N = 0.3$, 0.6, 0.9	fail, pass, pass	D	gpt-b
Maximum drift of N, E' in the disorder scans	3.56×10^{-7}	D	gpt-b
Pinned agents' solver against the independent full-tensor field (step 0.001)	4.9×10^{-9}	D	gpt-b
Proposer's toy prediction at $L_d \approx L_{\mathrm{NL}}$ (failed): $g^{(2)}$ (predicted 1.80)	1.74, 1.70, 1.67	T	claude-interactive.7
Scrambling and Gaussianity			
Exact one-mode Kerr, $n = 4096$, $s = 4$: coherence; $\det V$ (pure Gaussian 1/4)	0.99610; 2.51318	S	gpt-b
TWA half-collapse at $n = 10^8$: disorder path 7501 at steps 0.04 and 0.02; path 7502 at 0.04	331, 383; 411 L_{NL}	S	gpt-b
Phase memory and failure of RJ without disorder, $K = 5$: longest length	$1.2 \times 10^6 L_{\mathrm{NL}}$	G	gpt-b
Aligned launch without disorder: long-time $\langle I_0 \rangle$ against the refined conditioned-ensemble prediction	0.56881 ± 0.00193 vs 0.58127 ± 0.00204	G	gpt-b

Tangent maps are propagated analytically along the same path; TWA ensembles sample the Wigner function of the coherent input and share the quenched disorder path of their trajectory. Uncertainties cluster by disorder realization; time samples within a window are not counted as independent. Applicable artefact checks were step refinement (with nested Brownian bridges), trajectory number, Fock cutoff, mode truncation ($K = 4$), symplectic reciprocity, conservation of N and E' (drifts below 4×10^{-7}) and total-number shot noise. The temporal window, aliasing and Raman-grid checks do not apply to this monochromatic model. A second implementation, the agents' copy of FastDifferentiableMMNLO (fixed-step Float64 RK4IP) at commit `b9fa957`, agrees with the independent full-tensor integration to 4.9×10^{-9} in the field and, after a tensor-index convention was resolved, to 1.2×10^{-7} dB in the squeezing observables. Predictions and every extension were committed before the corresponding production runs; raw data are outside the repository at the locations recorded in each run's metadata.

Appendix G: Provenance of the numbers

Tables II and III list every number quoted in the text with the run that produced it and the agent that ran it; run ids are folders under `runs/` in the project repository. All dynamical

TABLE III. Provenance, continued (squeezing). Run abbreviations as in Table II.

Quantity	Value	Run	Agent
Early multimode squeezing			
λ_{lin} at $z = 10L_{\mathrm{NL}}$ and $20L_{\mathrm{NL}}$, all tested launches and disorder paths with $E'/\kappa N = 0.3$–0.6	−24.72 to −25.74 dB; −30.12 to −31.56 dB	Q	gpt-b
f_0 of the $E'/\kappa N = 0.9$ launch at $z/L_{\mathrm{NL}} = 5, 10, 20$	0.418, 0.415, 0.501	Q	gpt-b
Pinned agents' solver against the independent tangent observables	1.2×10^{-7} dB	Q	gpt-b
Curvature coefficient c ($E'/\kappa N = 0.6$, $z = 10L_{\mathrm{NL}}$, launch 4201, disorder 7601)	0.0920 ± 0.0039	Q	gpt-b
Occupation exponent of the excess, $n = 3 \times 10^5$–3×10^6	-0.9947 ± 0.0103	Q	gpt-b
Largest $\lvert 10 \log_{10}(V_{\mathrm{fixed}}/\lambda_{\mathrm{lin}})\rvert$ inside the envelope (67 cases)	0.366 dB	Q	gpt-b
Held out, $E'/\kappa N = 0.3$, $n = 10^8$: excess at $z/L_{\mathrm{NL}} = 10, 20, 50$	0.299, 6.743, 28.648 dB	Q	gpt-b
Lowest optimized quadrature, $n = 10^{10}$, 25 sampled distances	−31.487 dB	Q	gpt-b
Late counting discrepancy before/after the final step refinement	2.590 / 0.402 dB	Q	gpt-b
Exact single mode, $R = 0.8$: relative excess at $n_0 = 10^4$, 10^{10}	0.11060, 0.10015	K	gpt-b
Loss: 90% transmission and 90% detection; resulting ceiling	$\lambda_{\mathrm{det}} = 0.19 + 0.81\lambda$; −7.21 dB	C, Q	gpt-b
Condensate's own squeezing ($E'/\kappa N = 0.3$, $L_d/L_{\mathrm{NL}} = 11.1$, $n = 10^{10}$, 12 paths)			
Largest error of Eq. (8); largest V_X/V_P (1177 covariances)	0.062 dB; 0.023	C	gpt-b
λ_- at $z = 10L_{\mathrm{NL}}$: mean; SD; SE	−9.487; 1.651; 0.477 dB	C	gpt-b
$1/V_P$ at $z = 10L_{\mathrm{NL}}$, mean; f_0 range	−24.97 dB; 0.80–0.86 (fig.)	C	gpt-b
λ_- at $z = 20L_{\mathrm{NL}}$: all 12 paths; 8 paths with $f_0 \le 0.87$	-9.547 ± 2.906; -7.952 ± 1.978 dB	C	gpt-b
Largest $V_P^2/(8n_0\lambda_-)$ at $z = 10, 20L_{\mathrm{NL}}$	2.4×10^{-5}, 6.3×10^{-4}	C	gpt-b
Largest nonlinear-Wigner minus tangent difference, frozen ensembles, $z \le 20L_{\mathrm{NL}}$	0.175 dB	C	gpt-b
Largest per-path step change at $z = 50L_{\mathrm{NL}}$	1.288 dB	C	gpt-b
SPM reference ratio $(1 + 4\Phi_{\mathrm{self}}^2)/V_P$ at $z/L_{\mathrm{NL}} = 10, 20, 50$	0.822, 0.863, 0.889	C	gpt-b
Kerr stage, no-go, Sagnac			
Exact moments versus Fock evolution	3.93×10^{-12}	K	gpt-b
Largest coefficient error, $1 \le \nu \le 30$, $10^8 \le n_d \le 10^{12}$	0.412%	K	gpt-b
Exponent at $\nu = 10$; relative-error exponent	−0.334621; −0.328378	K	gpt-b
Largest error of the nine frozen held-out cases (cards: 0.218)	0.217 dB	K	gpt-b
Extracted condensate, 12 cases: largest |error| (cards: 0.176); case $z = 10L_{\mathrm{NL}}$, $n = 10^{10}$	0.175 dB; $\nu = 5.68$, −23.42 dB	C	gpt-b
Oriented Gaussian counterexample at $n_d = 10^{10}$: variance versus coherent optimum	10^{-4} vs 4.386×10^{-4}	K	gpt-b
Unimodal super-Poissonian counterexample: mean, variance	20.89, 220.10	K	gpt-b
Sagnac, $\Phi = 30$, 10^{10} per arm, bright reference: no jitter; Dirichlet pump (ε^2)	−35.565; −25.409 dB (2.7085×10^{-3})	K	gpt-b
Raw normalization error at $\varepsilon = 0.05099$, $\Phi = 3$; external-LO baseline at $\Phi = 30$	1.135%; −33.566 dB	K	gpt-b
Error of baseline plus ε^2 for the Dirichlet pump at $\Phi = 30$	3.609%	K	gpt-b

and exact-quantum numbers come from gpt-b's cross-family runs, except the failed proposer-family toy prediction, which is listed because it is part of the record.

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